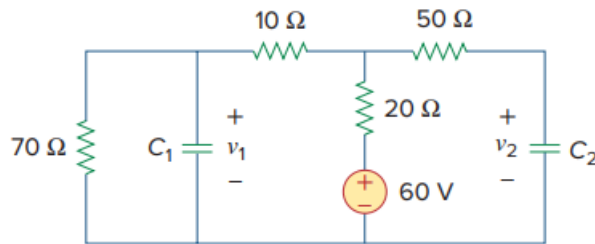


Problem

Find the voltage across the capacitors in the circuit of Fig. 6.49 under dc conditions.

Fig. 6.49:



Step-by-step solution

Step 1 of 3

Refer to the circuit in Figure 6.49 in the textbook.

Under DC conditions all the capacitors acts like as an open circuit as shown in Figure 1.

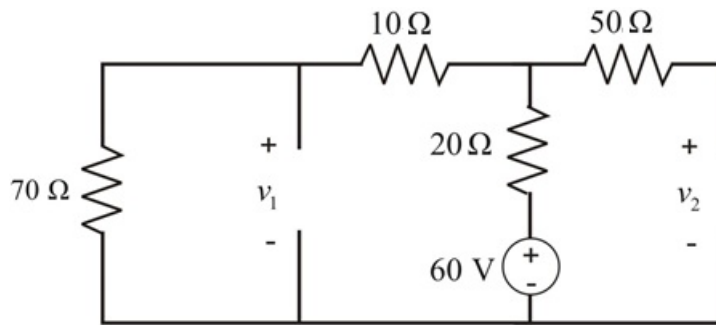


Figure 1

Step 2 of 3

Current through the 70Ω resistor is,

$$i_{70\Omega} = \frac{60}{70 + 10 + 20} \\ = 0.6 \text{ A}$$

The voltage across capacitor C_1 is,

$$v_1 = 70(i_{70\Omega})$$

Substitute 0.6 A for $i_{70\Omega}$.

$$v_1 = 70(0.6) \\ = 42 \text{ V}$$

Thus, the voltage across the capacitor C_1 under dc conditions is 42 V.

Step 3 of 3

The voltage across capacitor C_2 is,

$$v_2 = 60 - 20i_{70\Omega}$$

Substitute 0.6 A for $i_{70\Omega}$.

$$\begin{aligned} v_2 &= 60 - 20(0.6) \\ &= 48 \text{ V} \end{aligned}$$

Thus, the voltage across the capacitor C_2 under dc conditions is 48 V.